

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Backend, frontend, dataset, model, and docs are well organized.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are provided for important sections only.
6	Modular Design	Code is split into reusable functions/modules	Yes	React components and Flask modules are reusable.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unnecessary duplicate code has been removed.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	PredictionForm	React.js	Crop Prediction Page	High

2	ResultCard	React.js	Prediction Results	High
3	Navbar	React.js	Python / Flask	High
4	Flask Prediction API	Python / Flask	Backend Prediction Service	High
5	Random Forest Model	Scikit-learn	Crop Recommendation	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured project folders and modules.
Readability	5	Code is easy to understand and maintain.
Reusability	5	Reusable React components and backend modules.
Documentation / Comments	4	Comments are added where necessary.
Overall Score	5	Clean, modular, and maintainable project.